

Sai Kiran Mediga

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Professional Summary

- Having 5 years of experience as a **Web Application** Developer. Strong involvement in Analysis, Design, Development, Implementation, Testing, and support of different Web Applications utilizing **Python**, and **Django**.
- Good knowledge of various stages of SDLC (Software Development Life Cycle), Software Testing Life Cycle (STLC), and QA methodologies from project definition to post-deployment documentation.
- Experienced in using MVC architecture using **RESTful API**, and high-level Python Web frameworks like **Django** and **Flask**. Experienced object-oriented programming (**OOP**) concepts using Python, and Django.
- Hands-on experience in developing and designing a user interface using **HTML**, **CSS**, **JavaScript**, **Bootstrap**, **jQuery**, and **Ajax**. Good skills in JavaScript frameworks like **Reactjs**.
- Experience in developing high-performance web APIs using **FastAPI** for efficient JSON data consumption and proficiently testing APIs using **Postman** to ensure compliance with specifications and robustness in data handling.
- Good experience in developing **microservices** by using Rest API for data processing from databases like **SQL**, **MongoDB**, and **Cassandra**.
- Skilled in SQL for **Oracle**, **Postgres**, and **MySQL**, including writing queries, sub-queries, and functions. Proficient in SQL **Alchemy** for **ORM** and **Alembic** for database migrations, ensuring optimized data operations and consistent development environments.
- Experience in working with various Python IDEs like **Visual Studio code**, **PyCharm**, **Jupyter Notebook**, **Google Colab**, and **Sublime Text**, for deploying on cloud environments like **AWS**, **Heroku**, and **Microsoft Azure**.
- Proficient in implementing **JWT** (JSON Web Tokens) for secure authentication and authorization processes, for token generation and verification to enhance application security and streamline user management.
- Proficient in **CI/CD** pipeline design and implementation with **Jenkins**, **GitLab CI**, and **GitHub Actions**, automating integration, testing, and deployment. Experienced in integrating testing frameworks to ensure code quality and reliability.
- Proficient in **containerization** technologies like **Docker** and **Kubernetes**, enabling consistent and scalable deployment of applications across development, staging, and production environments.
- Proficient in writing unit testing code using **Unit Test**, and **PyTest**, and integrating the test code with the build process to validate the functionality of APIs and microservices.
- Expert in **AWS** services including **S3**, **EC2**, **Load Balancers**, and **Lambda** for scalable hosting and secure content management. Knowledgeable in AWS security, implementing **IAM** roles and policies for resource protection.

Skills

Languages	Python, SQL, JavaScript
Python Frameworks	Django, Flask, Rest framework,
JavaScript Framework	React.js, jQuery
Web Technology	HTML, CSS, JSON, Ajax, Restful, MVC, Bootstrap

Databases	HTML, CSS, JSON, Ajax, Restful, MVC, Bootstrap
IDE's & Tools	Vs Code, Sublime, jupyter notebook, Google colab, Pycharm, thonny.
Version Control tools	Git
Cloud Services	AWS, Microsoft Azure
CI/CD	Jenkins
Containerization Services	Docker, Kubernetes
Testing Tools	Pytest, Unit testing, Postman
Package Manager	Npm, pip, pipenv

Work Experience

Paychecx, (Remote) NJ Python Developer

Jan 2024 – Current

Responsibilities:

- Developed web applications with **Django**, utilizing its ORM for smooth database interactions and views for implementing business logic. Managed request handling efficiently with simple **URL routing** strategies.
- Developed scalable **RESTful APIs** with Django REST Framework, leveraging serializers for data serialization, **JWT** for authentication and secure access, and Django's ORM for efficient data querying, adhering to REST principles for enhanced service maintainability.
- Integrated **Postgres** schemas for user profiles, multimedia metadata, and access controls, optimizing database performance. Developed advanced queries and functions in **MySQL** and **PostgreSQL** to improve data manipulation efficiency.
- Developed a responsive front end utilizing **HTML**, **CSS**, and **JavaScript**, integrated with Django REST API by using **AJAX** for **asynchronous data retrieval** and manipulation, ensuring an interactive and seamless user experience.
- Implemented comprehensive testing strategies using Django REST Framework, **Pytest**, and **Postman** for **API validation**, employing unit, integration, and API client tests to ensure code quality and robust endpoints.
- Configured **Jenkins pipelines** for continuous integration and deployment of Django applications, leveraging **Kubernetes** for automated scaling and management of **containerized environments**.
- Deployed Django applications on **AWS EC2 instances**, utilizing **RDS** for managed database services and **S3** for static and media file storage, ensuring scalable and reliable hosting solutions.
- Integrated AWS services such as **Lambda** and API Gateway with Django REST API to develop serverless architectures, enhancing application scalability and reducing operational costs.
- Utilized **PyTest** to develop and execute comprehensive test suites encompassing unit, integration, and **functional testing**, ensuring thorough function validation and maintaining code integrity.
- Designed and implemented a **CI/CD** pipeline, automating code integration, testing, and deployment processes, and optimized **Dockerfile** and **YAML** configurations for efficient Docker images, deploying them on **AWS EKS** with **Kubernetes** for scalable and reliable application hosting.

Environment: Python, Django, MVC, AJAX, HTML, DOM, CSS, Bootstrap, JavaScript, JSON, REST, My-SQL, PostgreSQL, Jenkins, pytest and Postman.

Python Developer**Responsibilities:**

- Used **FastAPI** to develop high-performance backend services, facilitating efficient real-time data processing and exchange.
- Designed secure **authentication mechanisms** in FastAPI applications, utilizing **OAuth2** and **JWT authentication** services to secure user registration and login processes, ensuring data privacy access control and user authentication.
- Implemented **Alembic** for database migrations and version control in FastAPI projects, ensuring seamless schema updates and data integrity with **MySQL**.
- Integrated **Redis** as a caching and session store in FastAPI applications, optimizing performance and scalability by efficiently handling transient data.
- Established a comprehensive test suite with **PyTest**, focusing on functionality, integration, and performance testing to uphold high-quality standards.
- Implemented **CI/CD pipelines** using **Azure DevOps** for automated testing, building, and deployment of FastAPI applications on Microsoft Azure, ensuring streamlined development workflows and consistent delivery of high-quality software.
- Deployed FastAPI applications on **Microsoft Azure**, utilizing services like **Azure App Service**, **Azure Functions**, and Azure SQL Database to create scalable and resilient cloud-based architectures.

Environment: Python, FASTAPI, MySQL, Alembic, Redis, OAuth2, JWT, CI/CD pipeline, Microsoft Azure, pytest, and JWT.

Software Developer**Responsibilities:**

- Developed dynamic front-end interfaces using **HTML**, **CSS**, and **JavaScript**, integrated with **Flask** backend for efficient server-side rendering and **AJAX** for asynchronous data fetching.
- Developed a **microservice-based** backend architecture using Flask, employing **Flask-Restful** for constructing RESTful APIs and integrating **Flask-Caching** for enhanced performance, resulting in a scalable and efficient system for web application development.
- Utilized Flask with **SQLAlchemy** to interact with **PostgreSQL**, implementing robust database operations and ORM techniques for efficient data management and retrieval in web applications.
- Utilized **Flask-Login** and **Flask-JWT-Extended** for implementing secure user authentication and token-based authorization.
- Leveraged Django's ORM with PostgreSQL, utilizing raw SQL queries, query expressions, and Q objects for optimized database access and manipulation.
- Implemented robust permission checks in Django, including object-level permissions and custom permission classes, to ensure secure database interactions and maintain data integrity.
- Configured JWT expiration and refresh mechanisms, enhancing security by limiting the validity of tokens and requiring periodic renewal for continued access.
- Conducted comprehensive testing of Flask applications using **Unit testing** and **Flask testing**, implementing **unit** and **integration tests** to ensure code reliability and functionality in real-time scenarios.

Environment: Python, Flask, AJAX, REST API, JSON, SQL Alchemy, PostgreSQL, Agile, and unit test.

Education

Master of Computer Science

Bachelors